

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458025.8281 +/- 0.0037 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.209 +/- 0.062
Transit depth (Rp/Rs)^2	4.39 +/- 2.58 [%]
Orbital Inclination (inc)	87.56 +/- 1.88
Airmass coefficient 1 (a1)	0.4 +/- 0.25
Airmass coefficient 2 (a2)	0.74 +/- 0.52
Scatter in the residuals of the lightcurve fit is	66.01 %
Best Comparison Star	#1 - [627, 111]
Optimal Aperture	0.75
Optimal Annulus	3.0
Transit Duration (day)	0.085 +/- 0.019

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R fit vs published	0.209 ± 0.062 vs 0.1592 (deviation 0.53σ)
Duration fit vs published	0.085 d vs 0.0946 d (deviation 0.33σ)
Mid-time O-C	-7.7 min
Depth SNR	2.2
Residual scatter	66.01 %

Light curve

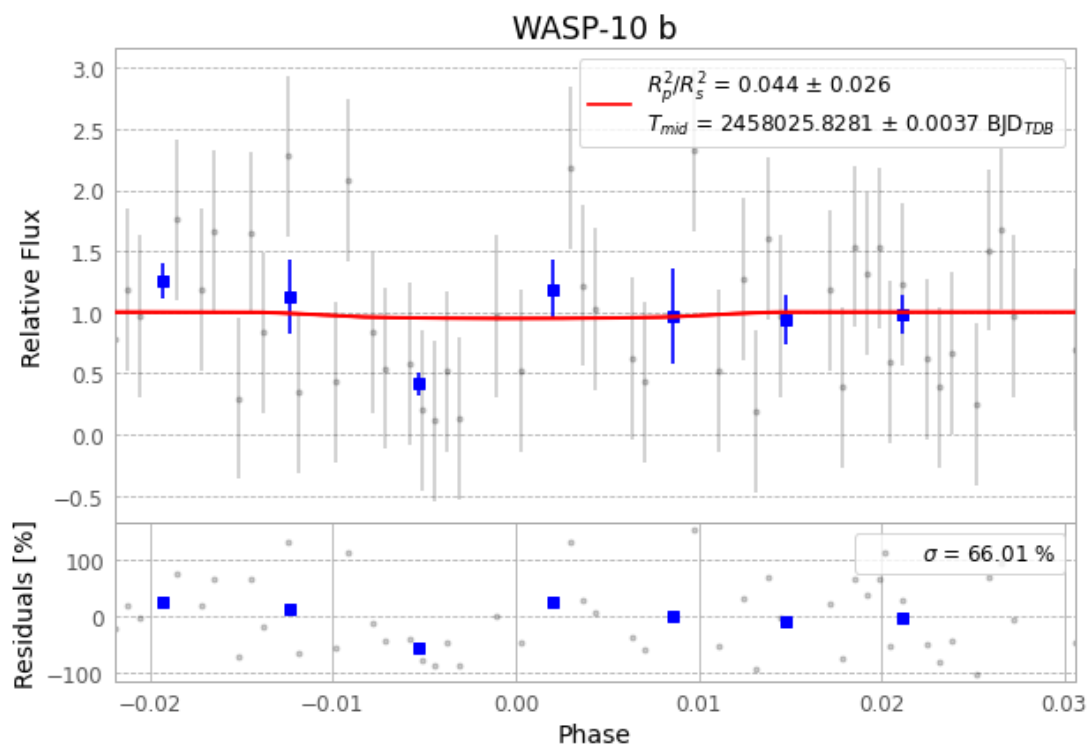


Figure 1 — FinalLightCurve_WASP-10 b_2017-09-28.png (SHA-256 b830bcaa913715ff...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [255, 88], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 ee0c8fa11a09030f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

83 FITS frames (55 MB), WASP-10170929055712.FITS → WASP-10170929100312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-170929.log (8abb909e8240...), FinalLightCurve_WASP-10 b_2017-09-28.png (b830bcaa9137...), FinalLightCurve_WASP-10 b_2017-09-28.csv (a9603e0004fb...)

Compute resources

Wall time 130.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 21:32Z.